

CPE 301: Embedded Systems Design Final Project Part 3

Team Members

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1. Project Motivation/Introduction

Our final project is the creation of a Scream-Activated Motor System. This system is designed to respond to sudden, high-decibel auditory inputs—such as a scream or a loud clap—by triggering an immediate mechanical and auditory response. It relies on a sound sensor to monitor ambient noise and compare it against a predefined threshold. When the threshold is breached, the system springs into action.

The motivation behind this project is to create an interactive device that can serve as a sudden-noise alarm, an interactive art installation, or an auditory stress-relief desk toy. By taking a simple input (loud noise) and routing it through an embedded system to control complex outputs (a servo motor, an active buzzer, visual LEDs, and an LCD display), this project effectively demonstrates the core principles of sensor integration, state machine logic, and actuator control.

2. Proposed System

2.1 System Schematic Diagram

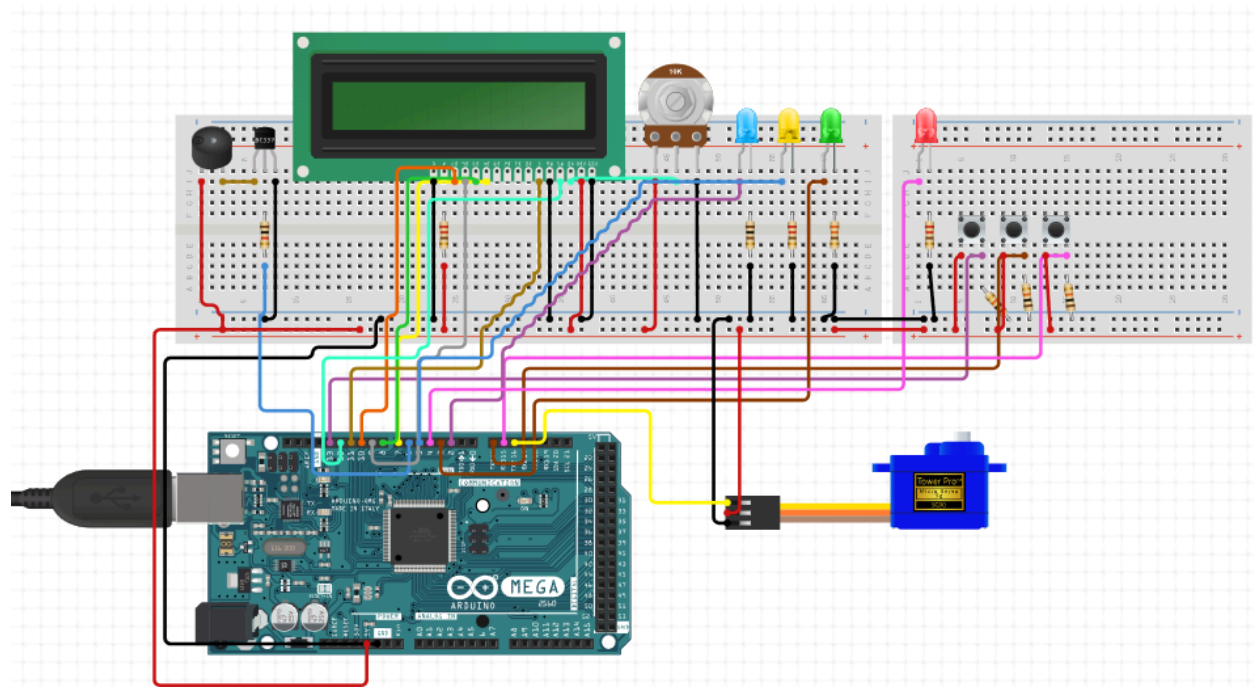


Figure 1: Online CAD of the Circuit (does not reflect actual wiring locations)

(Created by Qizhi Ge)

2.2 Detailed Component List

- Microcontroller: Arduino Mega 2560 (or standard Arduino) – The core processor handling ADC registers, state logic, and output control.
- Input Module: Sound Sensor Module – Detects ambient audio levels and provides an analog/digital signal to the microcontroller to trigger the threshold.
- Actuator 1: SG90 Micro Servo Motor – Rotates/spins when the system enters the active state.
- Actuator 2: Active Buzzer – Emits an audible alert when the noise threshold is exceeded.
- Display: I2C LCD Display (e.g., 16x2) – Displays the current operating state of the system in real-time (OFF, IDLE, ACTIVE, ERROR).
- Visual Indicators: 4 Distinct LEDs (e.g., Red, Green, Blue, Yellow) – Each LED uniquely maps to one of the four system states.
- Buttons (ON/OFF/RESET): 3 Different buttons distinguished by the color of the wire (Green, red, yellow) corresponding with on, off, and Reset.
- Miscellaneous: Breadboard, jumper wires, and current-limiting resistors for the LEDs.

3. System Behavior

3.1 Module Design

The system is divided into several interacting modules:

- Input Sensor Module: The sound sensor continuously monitors audio levels. It triggers an interrupt or registers a state change when the noise exceeds the set threshold.
- Visual & Text Output Module: The LCD display continuously updates to show the current state text ("OFF", "IDLE", "ACTIVE", "ERROR"), while the 4 LEDs serve as immediate visual state indicators.
- Action Module: The SG90 servo motor and the active buzzer wait for the "ACTIVE" state. Once triggered, the motor spins and the buzzer sounds to provide physical and auditory feedback.

3.2 Control Logic (State Diagram & state machine)

The system relies on a state machine with 4 distinct states:

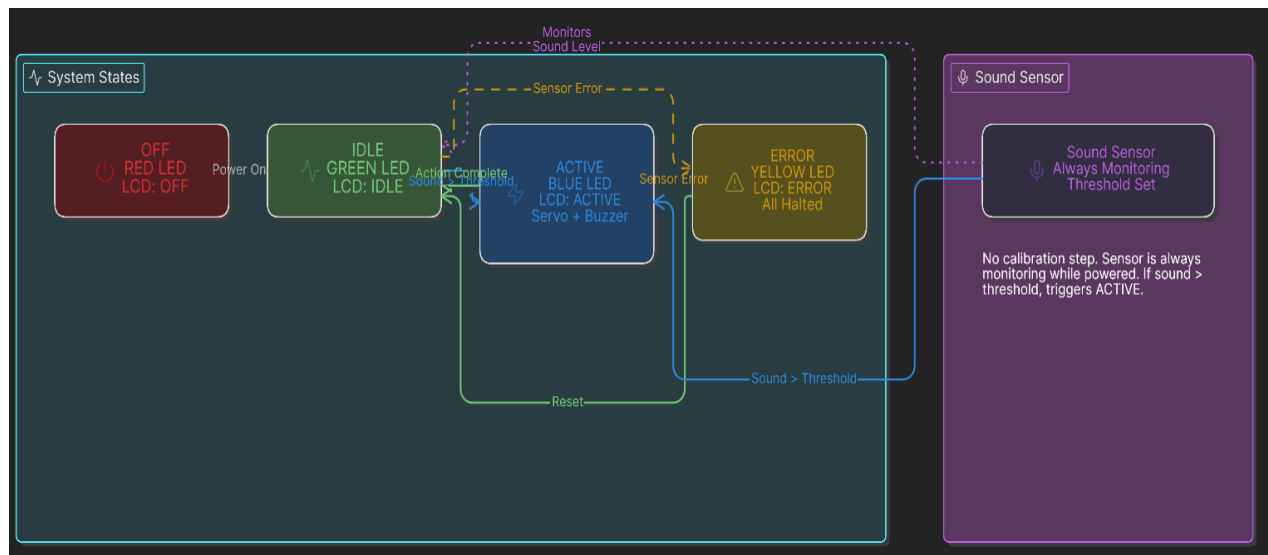


Figure 2: State Diagram of the System(created by Qizhi Ge)

1. State 1 (OFF): System is in standby. Actuators are disabled. The RED LED is on, and the LCD displays "OFF".
2. State 2 (IDLE): The system is actively listening, but the noise threshold has not been met. The GREEN LED is on, and the LCD displays "IDLE".
3. State 3 (ACTIVE): Triggered when the sound sensor detects a noise above the threshold. The BLUE LED is on, the LCD displays "ACTIVE", the SG90 servo motor spins, and the active buzzer sounds. After the action is complete, it returns to IDLE.
4. State 4 (ERROR): Triggered if a sensor disconnects or an invalid condition is met. The YELLOW LED is on, the LCD displays "ERROR", and all actuators are halted until the system is reset by the reset button.

Environmental:

Energy Efficiency: The system is designed to minimize overall power consumption by utilizing an efficient state machine architecture. Rather than running the SG90 servo motor and active buzzer continuously, these high-draw actuators remain powered down during the standby (OFF) and listening (IDLE) states. The system only draws peak current during the brief ACTIVE state when the noise threshold is breached, significantly reducing wasted energy over time.

Design Safety: Safety is prioritized through the use of low-voltage DC power (5V), which eliminates the risk of harmful electric shock to the user. On the hardware side, current-limiting resistors are properly utilized for all indicator LEDs to prevent component burnout and potential overheating. Additionally, the software features a dedicated "ERROR" state with visual feedback (Yellow LED) and a halt on all actuators, ensuring the system safely locks down if a sensor disconnects or an invalid condition occurs.

Affordability: The prototype relies exclusively on highly accessible, off-the-shelf microelectronics (such as the Arduino Mega, standard sound sensor, and SG90 servo). By utilizing these ubiquitous and cost-effective components, the project proves that reliable safety and monitoring systems can be manufactured affordably without compromising on the quality of the detection or response.

Sustainability: To reduce electronic waste, the current prototype is built entirely on a solderless breadboard. This ensures that every jumper wire, LED, resistor, and microcontroller can be completely disassembled, salvaged, and reused for future projects. Furthermore, by designing the system to be modular, individual parts (like upgrading to a larger sound sensor) can be swapped out and repaired without needing to discard the entire unit.

Accessibility: The system is designed with an inclusive, user-friendly interface that requires no technical expertise to understand. By providing simultaneous visual feedback (through a clear LCD text readout and color-coded LEDs) alongside auditory feedback (the active buzzer), the system ensures its operational state is immediately accessible to users, including those who may be hard of hearing or visually impaired. The hardware layout is also straightforward, making it easy to operate via the basic start/reset/off buttons.

Conclusion

System report: the system worked perfectly as intended, correctly showing the 4 states (off, idle, active, Error), and the Buttons (start/off/reset) also worked perfectly. Furthermore, the system was able to detect sound until a sound amplitude threshold was reached, at which point the system activated and actuated the servo motor to spin 90 degrees and displayed the active state on the LCD. Unfortunately, our current system has one big limitation: it will not activate unless the sound is directly next to the sound sensor. This can cause our system to not function as required due to the distance bottleneck caused by our relatively small sound detector, but this also means that it can be solved by adding a bigger speaker that is able to pick up sound relative to distance. This fix will be able to solve our only bottleneck because with the addition of a bigger sound sensor, we will no longer be hindered by distance and be able to implement this system anywhere as needed.

References.

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